ScienceHub — Locked Refinement

Personal Study OS • Refined Master Specification

Status: Locked refinement baseline for future ScienceHub development.

Core principle: Learn → Understand → Recall → Practice → Performance → Weakness → Revision → Mastery → Next Action.

1. Product Identity & Vision

- **ScienceHub** is a personal-use Study OS designed around the user's academic journey, with Class 11 as the primary focus and Class 12 integrated where concepts overlap or expand.
- The system is unified rather than a collection of disconnected mini-apps. Syllabus, Notes, Practice, PYQs, Exams, Performance, AI, Revision, PCB Stream, My Space and Search share the same underlying academic context.
- Primary educational subjects: Biology, Physics, Chemistry, English and Hindi.

2. Home / Hero — Locked Visual Direction

- Hero hierarchy: ScienceHub identity → prominent live DNA → mission statement → Enter ScienceHub → global search → supporting navigation.
- Primary visual identity is a prominent **3D genetic double helix**; the main hero does not use a brain as its central symbol.
- Mission language: “Your Study. Your Science. Your Future.” with the supporting idea “Learn • Practice • Understand • Grow.”
- The visual atmosphere combines personal study warmth with scientific discovery: student/study ambience is secondary, nature/discovery energy is atmospheric, and controlled gold is an accent rather than the primary logo color.

3. Live ScienceHub DNA Icon

- Central 3D DNA double helix, vertical and visually dominant (approximately 70–75%).
- DNA rotates around its own vertical/axial axis at **360° every 3 seconds** (120°/second).
- Exactly **6 electrons**. Each electron has its own separate long circular/elliptical 3D orbital path and a thin trailing line.
- The six orbital systems use differently oriented 3D planes so all six remain visually distinguishable. Orbital lines stay subtle; DNA remains dominant.
- Color language: deep blue/violet scientific glow, controlled cyan highlights, subtle gold accents and optional biological green highlights.
- The installed Android/PWA icon remains static; the live animation is used inside the application.

4. Syllabus Architecture

- Hierarchy: Class → Subject → Unit/Block → Chapter → Topic → Sub-topic → Concept → Practice → PYQ/Exam → Performance → Revision.
- Class 11 receives priority; Class 12 is connected where it extends or overlaps Class 11 learning.
- Each topic can move through: Not Started → Learning → Practicing → Revised → Mastered, with optional strength states such as Weak / Normal / Strong.

- Syllabus acts as the academic backbone connecting Notes, Search, Practice, PYQs, Exams, Performance, Revision, My Space and AI. Opening a chapter alone does not count as meaningful completion.

5. Practice Engine

- Practice flow: concept → questions → performance → weakness detection → targeted revision.
- Includes chapter, topic, weak-topic, mixed, PYQ, competitive, quick, focus, exam-simulation and recall practice.
- Question formats include MCQ, multiple-select, true/false, numerical, assertion/reason, matching, sequence, fill-in, short answer and diagram/image-based questions.
- **Locked scoring:** correct/good = **+2**; incorrect/bad = **–1**.
- Every question maps to Subject → Class → Chapter → Topic → Concept. Results track accuracy, attempts, mastery, weakness, speed and revision need.

6. PYQ & Exam Learning Pipeline

- PYQs begin from **2020 onward**, with tags for year, exam, syllabus status (including over-syllabus where applicable), subject, chapter, topic and concept.
- PYQs should follow concept learning rather than being an isolated archive.
- Exam tracking includes school and competitive contexts, with a broad All-India exam tracker and the user's requested distinction between exams that went with trouble and exams that went smoothly.

7. AI System

- **KuroVen:** default action-taking AI; starts work and converts intention into concrete next actions.
- **Hikaitage:** learning and strategic problem-handling AI; explains where/what/how/why/when with an experienced-teacher style.
- **HukoVaige:** psychology/behavior-analysis role focused on patterns, curiosity, growth, self-understanding and honest reflection.
- **Sukoon.Brain:** calm-support chatbot using Pause → Express → Understand → Organize → Next Small Step. It is not a therapist or medical replacement.
- **Aui / WORLD AI:** a movable world-knowledge assistant triggered when the user calls/types “Aui”; it is not a primary permanent role.

8. PCB Stream & Scientific Sticker System

- PCB Stream is a dedicated connected future-pathway module, not a subject badge. Its PCB visual identity can appear contextually across careers, courses, research, bioinformatics, scholarships, entrance exams and opportunities.
- PCB opportunities are placed at the final section of the overall ScienceHub structure.
- Scientific stickers are searchable reusable SVG assets with metadata, categories, synonyms and usage contexts. Coverage includes Biology/medical/genetics/biotechnology, Physics, Chemistry, equipment, nature/environment and study/concept stickers.
- Search supports related concepts and aliases; unknown terms should not be guessed. The sticker system is functional and connected to Notes, Concept Maps, Practice, Study Plans, PCB Stream and AI.

9. Search, Data, Privacy & Sharing

- Global powerful search belongs on Home; a smaller contextual search remains available for Notes and other relevant areas.

- User Data should support detailed study/time tracking: what was studied, how long, where and in which context, while avoiding unnecessary scope.
- Native device Share should support notes, questions, resources, concept maps, study plans and results with privacy controls and no unintended private-data sharing.
- Privacy boundaries should be explicit, especially for AI context and Sukoon.Brain conversations.

10. My Space, Settings & Device Behavior

- My Space contains detailed personal-study options; the **Recovery** box is the final option.
- Camera mode setting is included.
- When ScienceHub is open, the requested no-call/notification behavior hides notification/ring interruptions while keeping the ScienceHub session open, subject to device/OS limitations.
- Design target: mobile-first, app-like, offline-capable where technically feasible, and free/no-payment preferred.

11. Architecture Refinement & QA

- Core modules must share one consistent data model and navigation system.
- Prevent duplicate Home/mission renderers, mismatched states/statistics, multiple competing icon systems, duplicated Perspective systems and database/version migration conflicts.
- Final QA must cover startup, routing, assets, mobile responsiveness, Home/hero, study/data, AI routing, PCB integration, privacy, native sharing, offline/PWA behavior, performance and regression testing.
- Recommended refinement order: **Learning Engine** → **Recall/Revision** → **Performance/Mistakes** → **PYQ Engine** → **AI Orchestration** → **Global Search** → **Concept Map/Flashcards** → **School/Daily/Weekly/Monthly** → **Exam Lifecycle** → **Data/Privacy/Share** → **final architecture QA.**

12. Locked Core Philosophy

- ScienceHub should not merely store study material. It should understand the user's academic structure and turn learning activity into a continuous improvement loop.
- The final system should preserve future compatibility: new modules, AI roles, exams, stickers, subjects, data fields and UI improvements should be addable without breaking the core architecture.

Locked Visual / Asset Baseline

Area	Locked baseline
Branding	sciencehub-logo.svg • sciencehub-wordmark.svg • sciencehub-icon.svg • sciencehub-mark.svg
Hero	dna-helix.svg • dna-helix.webp • hero-background.webp • hero-glow.webp
Subjects	biology.svg • physics.svg • chemistry.svg • english.svg • hindi.svg
PCB	pcb-badge.svg + searchable scientific sticker library
UI	home/search/notes/practice/exams/settings + loading/empty/error states
Stickers	pcb/physics/chemistry/equipment/nature/study/universal asset families

Document note: This PDF records the current locked refinement baseline. Future changes should be treated as explicit refinements rather than silently altering locked behavior.